Forecasting Stock Market Volatility Using Daily S&P 500 ETF Returns

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1 Introduction, Motivation, and Objectives

Financial market uncertainty is often expressed through volatility rather than price levels alone. For many practical applications, including risk management, portfolio allocation, derivative pricing, and stress testing, the path of volatility is more informative than the path of prices by itself. Daily equity returns are well known to exhibit volatility clustering, in which large shocks tend to be followed by additional large shocks, while calm periods tend to persist. This feature makes financial volatility a natural target for time series modeling.

In this project, we use the SPDR S&P 500 ETF Trust (SPY) as a liquid proxy for the U.S. equity market and evaluate whether its short-term volatility can be forecasted from historical information. SPY is attractive for this purpose because it is highly traded, spans multiple market regimes, and reflects broad market behavior rather than the idiosyncratic dynamics of a single stock. The project therefore combines a substantively meaningful financial question with a time series structure that is well suited to the models covered in class.

The analysis is organized around three research questions. First, can short-term stock market volatility be forecasted using standard time series models? Second, do more advanced models outperform simple persistence-based baselines? Third, which model provides the best out-of-sample accuracy when evaluated on a held-out test period? The final report addresses these questions using a common volatility proxy, multiple forecasting models, and two complementary evaluation schemes.

GitHub repository: Pat-Zhu/TSA_Final_Project

2 Dataset Information and Preprocessing

Daily SPY price data were obtained from Yahoo Finance. Adjusted closing prices were used to compute daily log returns, and volatility was then defined as the 21-day rolling standard deviation of returns. This choice was motivated by both interpretation and practicality. A 21-trading-day window is close to one trading month, so the resulting series captures short-run market risk while smoothing some of the day-to-day noise in returns. At the same time, the window is short enough to preserve abrupt changes in market conditions.

After differencing and rolling-window calculation, missing values were removed and the cleaned data were arranged chronologically. The final dataset contains 2805 observations spanning 2015-02-03 to 2026-03-30. The series was divided into an 80% training period and a 20% test period. The training set contains 2244 observations and ends on 2024-01-02, while the test set contains 561 observations and begins on 2024-01-03. This chronological split preserves the forecasting setting and avoids information leakage.

The cleaned dataset contains four variables used throughout the report: date, adjusted close, daily log return, and the 21-day rolling volatility proxy. The summary table below provides basic descriptive statistics, and the following table shows the first ten rows of the processed series.

Table 1: Summary of variables used in the analysis.

Variable	Description	Unit	Mean	SD	Min	Max
date	Trading date	Date	NA	NA	NA	NA
adj_close	SPY adjusted close	USD	345.5877	147.1009	154.5588	693.5958
return	Daily log return	Log return	0.0005	0.0112	-0.1159	0.0999

Variable	Description	Unit	Mean	SD	Min	Max
volatility_21	21-day rolling return volatility	Standard deviation	0.0094	0.0062	0.0022	0.0590

Table 2: First 10 rows of the cleaned dataset used in this analysis.

	date	adj_close	return	volatility_21
22	2015-02-03	169.6365	0.0144	0.0113
23	2015-02-04	168.9905	-0.0038	0.0105
24	2015-02-05	170.6965	0.0100	0.0105
25	2015-02-06	170.2244	-0.0028	0.0102
26	2015-02-09	169.4625	-0.0045	0.0095
27	2015-02-10	171.2678	0.0106	0.0096
28	2015-02-11	171.3672	0.0006	0.0094
29	2015-02-12	173.0153	0.0096	0.0095
30	2015-02-13	173.7274	0.0041	0.0094
31	2015-02-17	174.0007	0.0016	0.0090

2.1 SPY Price Dynamics

The adjusted closing price series shows the long-run growth of the broad U.S. equity market but also reveals pronounced drawdowns during major stress episodes. This plot is useful mainly for contextualizing the return and volatility series analyzed later.

2.2 Daily Log Returns

The daily return series fluctuates around zero, as expected, but the magnitude of the fluctuations varies substantially over time. Large positive and negative returns cluster in particular periods, suggesting time-varying variance rather than constant noise.

2.3 Rolling Volatility Proxy

The rolling volatility series below is the main target of the forecasting exercise. It smooths raw returns while preserving the large spikes associated with market stress. The most obvious feature is volatility clustering: elevated volatility does not occur in isolation but rather persists for stretches of time.

2.4 Distributional Features of the Volatility Proxy

The volatility proxy is strictly nonnegative and right-skewed. Most days are concentrated at relatively low volatility levels, while a smaller number of days account for the large spikes visible in the time series plot. This asymmetry is typical of financial risk measures and helps explain why forecasting large episodes is more difficult than forecasting ordinary days.

Figure 1: Adjusted closing price of SPY over the sample period.

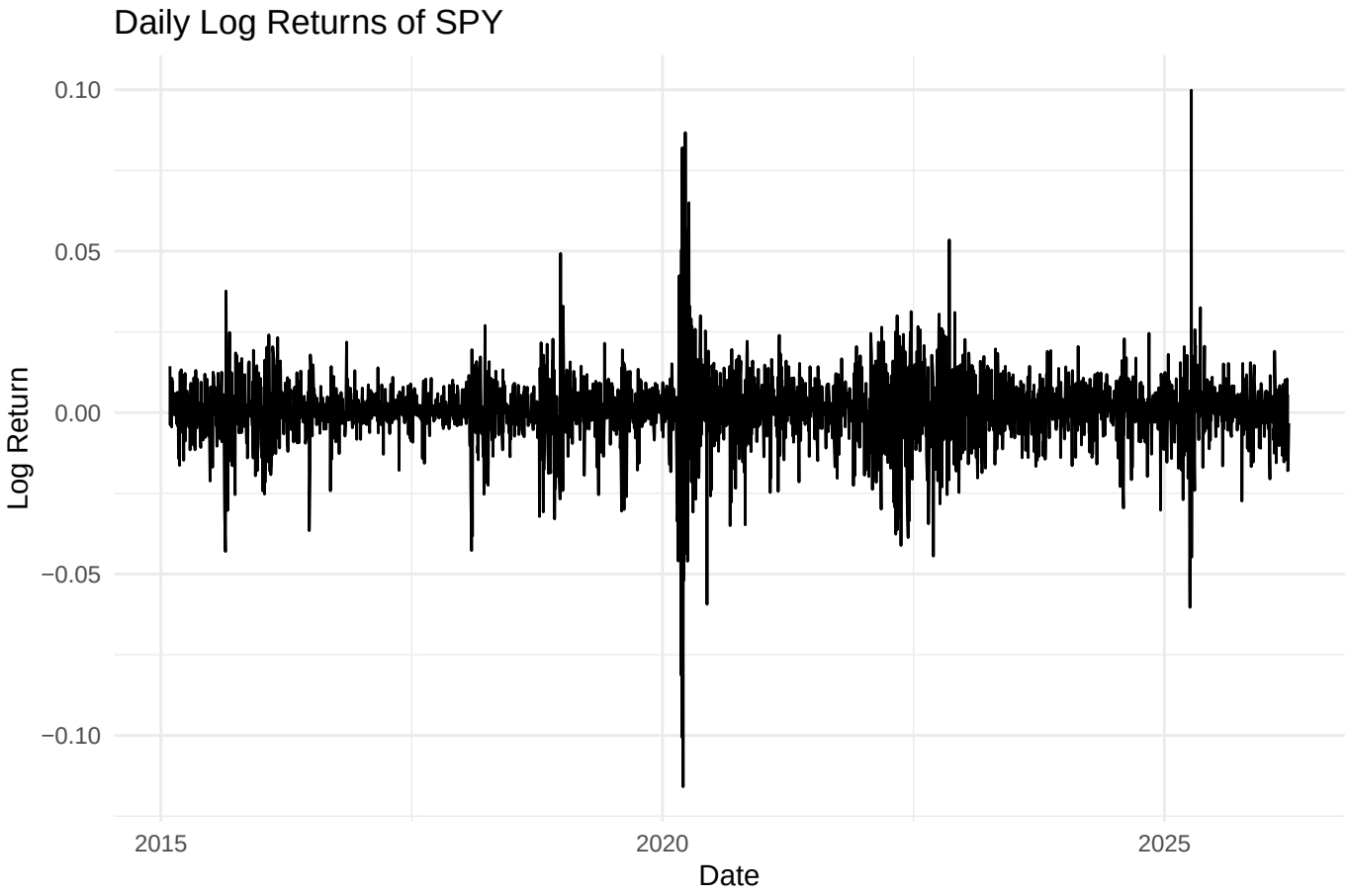

Figure 2: Daily log returns of SPY.

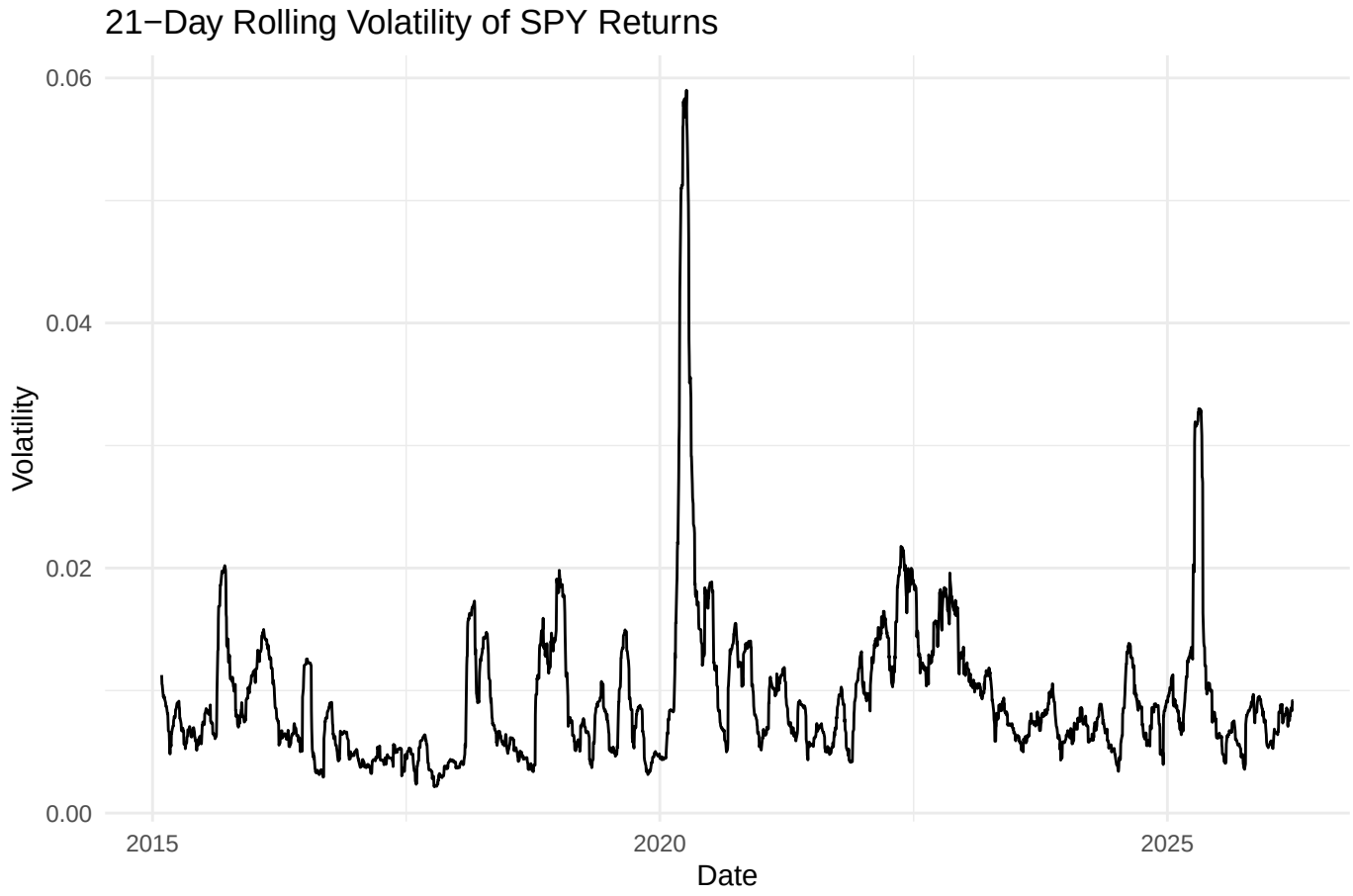

Figure 3: Twenty-one-day rolling volatility of SPY daily log returns.

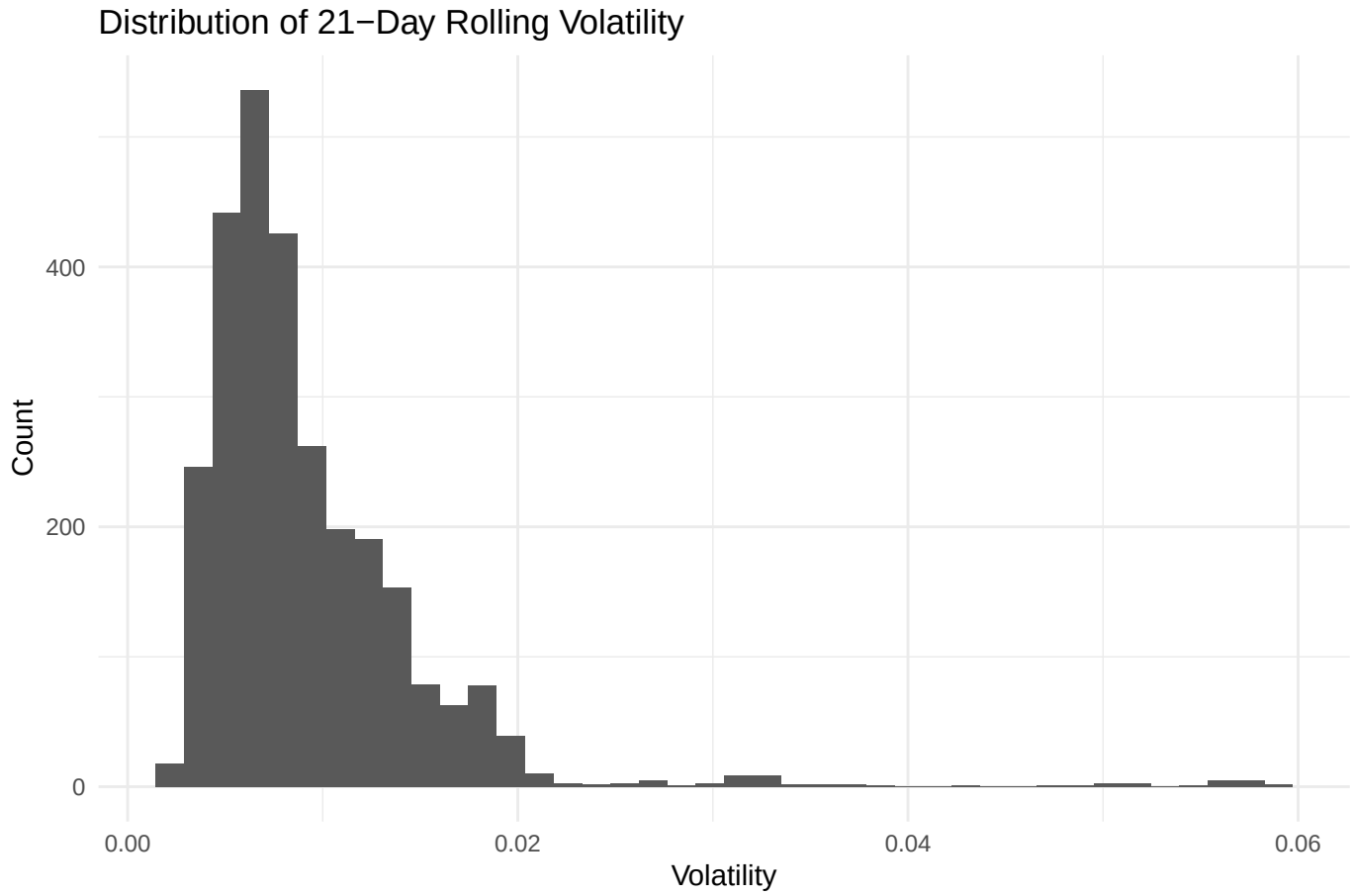

Figure 4: Distribution of the 21-day rolling volatility series.

3 Time Series Characteristics and Evaluation Design

The plots above suggest that the volatility proxy is persistent, heteroskedastic, and far from white noise. Before fitting forecasting models, it is useful to characterize that dependence more formally and to explain how out-of-sample forecasts will be evaluated.

3.1 Training-Test Split

The figure below shows the training and test periods on the same volatility series. The split was chosen so that the models are trained on a long historical window but still evaluated on a substantial, recent holdout period containing meaningful variation.

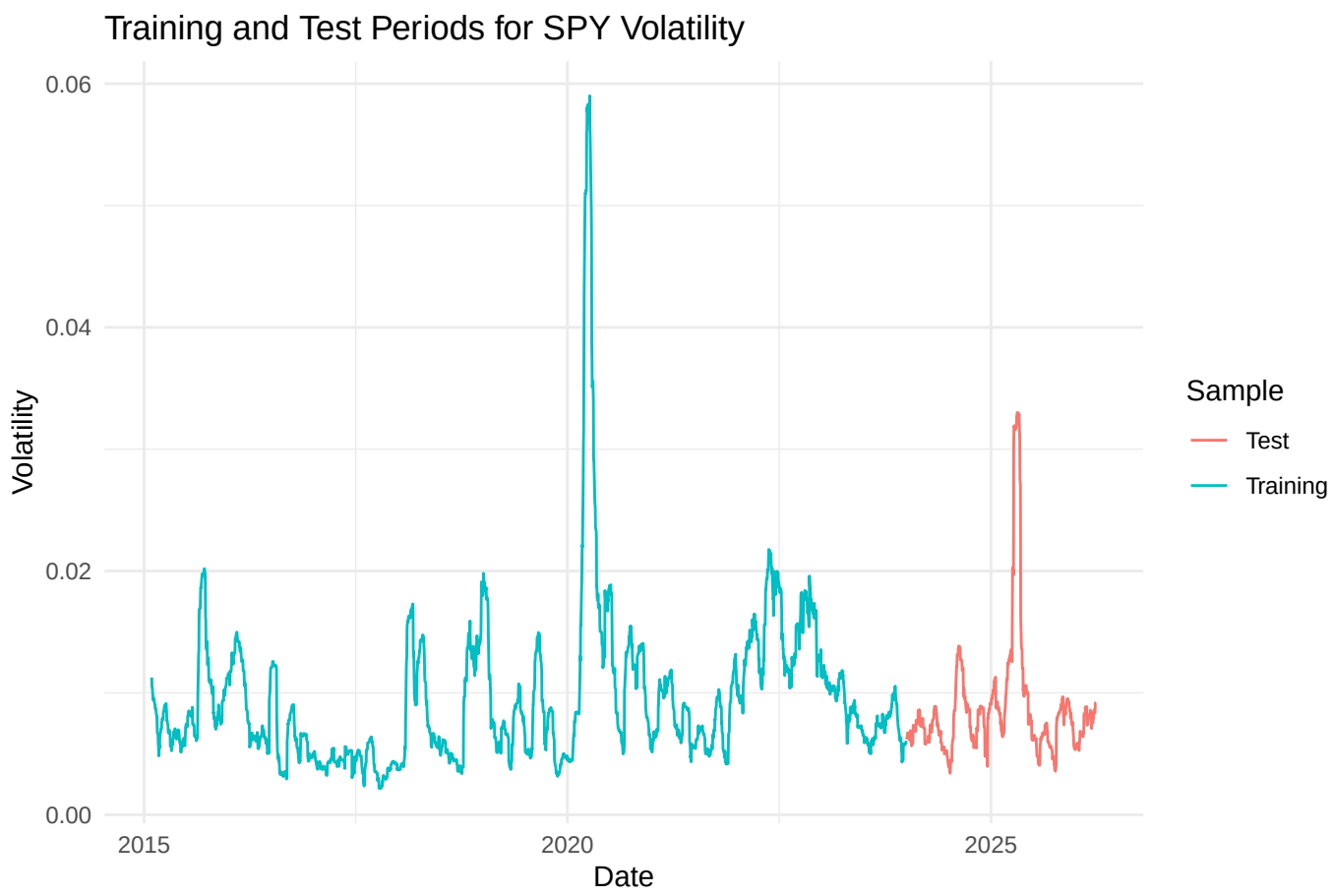

Figure 5: Training and test split for the volatility series.

3.2 Autocorrelation Structure

The autocorrelation function (ACF) and partial autocorrelation function (PACF) provide an initial view of persistence in the volatility series. The ACF displays strong short-lag dependence and decays gradually, while the PACF shows a dominant first lag and smaller early-lag effects. These patterns are consistent with a highly persistent process and suggest that recent volatility contains useful information for forecasting future volatility.

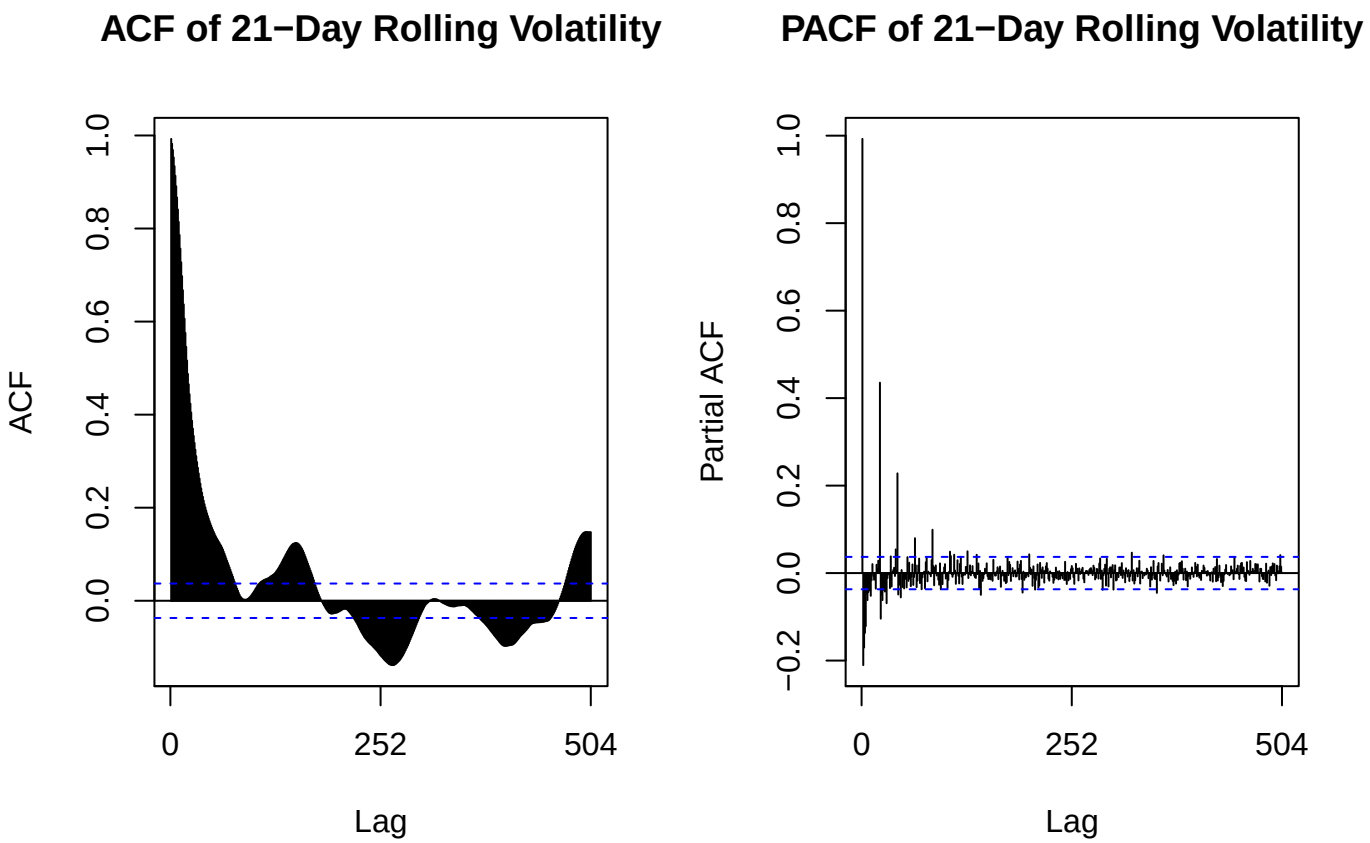

Figure 6: Autocorrelation (ACF) and partial autocorrelation (PACF) of the volatility series.

3.3 Preliminary Statistical Tests

To supplement the visual diagnostics, an augmented Dickey-Fuller test and a Ljung-Box test were applied to the training volatility series. The ADF test evaluates whether the series contains a unit root, while the Ljung-Box test evaluates whether a group of autocorrelations is jointly zero. Together, these tests help clarify whether time dependence remains substantial enough to motivate forecasting models beyond white-noise assumptions.

Table 3: Preliminary tests applied to the training volatility series.

Test	Statistic	P_Value
ADF test on training volatility	-6.6723	0.01
Ljung-Box test on training volatility (lag = 20)	29639.2901	0.00

The ADF test produced a p-value of 0.0100, while the Ljung-Box test produced a p-value of 0.0000. Regardless of the exact stationarity interpretation, the dependence structure is clearly strong enough to justify dynamic forecasting models, and the Ljung-Box result confirms that serial dependence is not negligible.

3.4 Two Evaluation Schemes

Model evaluation depends not only on the selected model but also on the forecasting design. Two schemes are therefore used in this analysis.

The first is a **single-origin benchmark**. Under this design, each model is fitted once to the training data and then asked to forecast the entire test horizon. This is a common benchmark and keeps the analysis consistent with the presentation workflow. However, for long test windows, simple models such as Naive and ETS often produce visually flat forecast paths because they are optimized for point prediction rather than dynamic tracking over hundreds of periods.

The second is an **updating one-step-ahead evaluation**. Under this design, each model produces a forecast for the next test observation only. Once that observation is realized, it is appended to the available history and the next one-step forecast is generated. This design is more informative for visual path comparison and is especially appropriate for volatility, because it reflects how a forecaster would update beliefs over time as new information arrives.

For the GARCH model, the implementation required special care. GARCH is fitted to the return series rather than to the rolling volatility proxy directly. In the single-origin benchmark, the conditional variance forecast is propagated forward analytically. In the updating one-step-ahead evaluation, the conditional variance is updated recursively using the realized lagged return and the previous conditional variance estimate. This change corrects the earlier flat-line implementation and aligns the report with the model behavior described in the presentation.

$$\hat{\sigma}^2_{t+1|t} = \omega + \alpha\varepsilon_t^2 + \beta\hat{\sigma}^2_{t|t-1}$$

where ω, α, and β are the estimated GARCH(1,1) parameters, ε_t is the most recent return innovation, and $\hat{\sigma}^2_{t|t-1}$ is the previous conditional variance estimate.

4 Individual Model Results

This section summarizes each model separately before comparing them collectively. For each model, the figure emphasizes the updating one-step-ahead path, because that representation is the most useful for assessing how the model behaves through time.

4.1 Naive Benchmark

The naive model uses the most recent observed volatility value as the forecast for the next period. In notation, the model is simply

$$\hat{y}_{t+1|t} = y_t.$$

Although extremely simple, this benchmark is important because volatility is persistent. A strong naive forecast therefore indicates that much of the forecasting problem can already be explained by local persistence alone.

In the updating one-step-ahead evaluation, the naive model tracks the general level of volatility reasonably well, but it cannot anticipate regime shifts or large changes in amplitude. Its main strength is stability and interpretability rather than responsiveness.

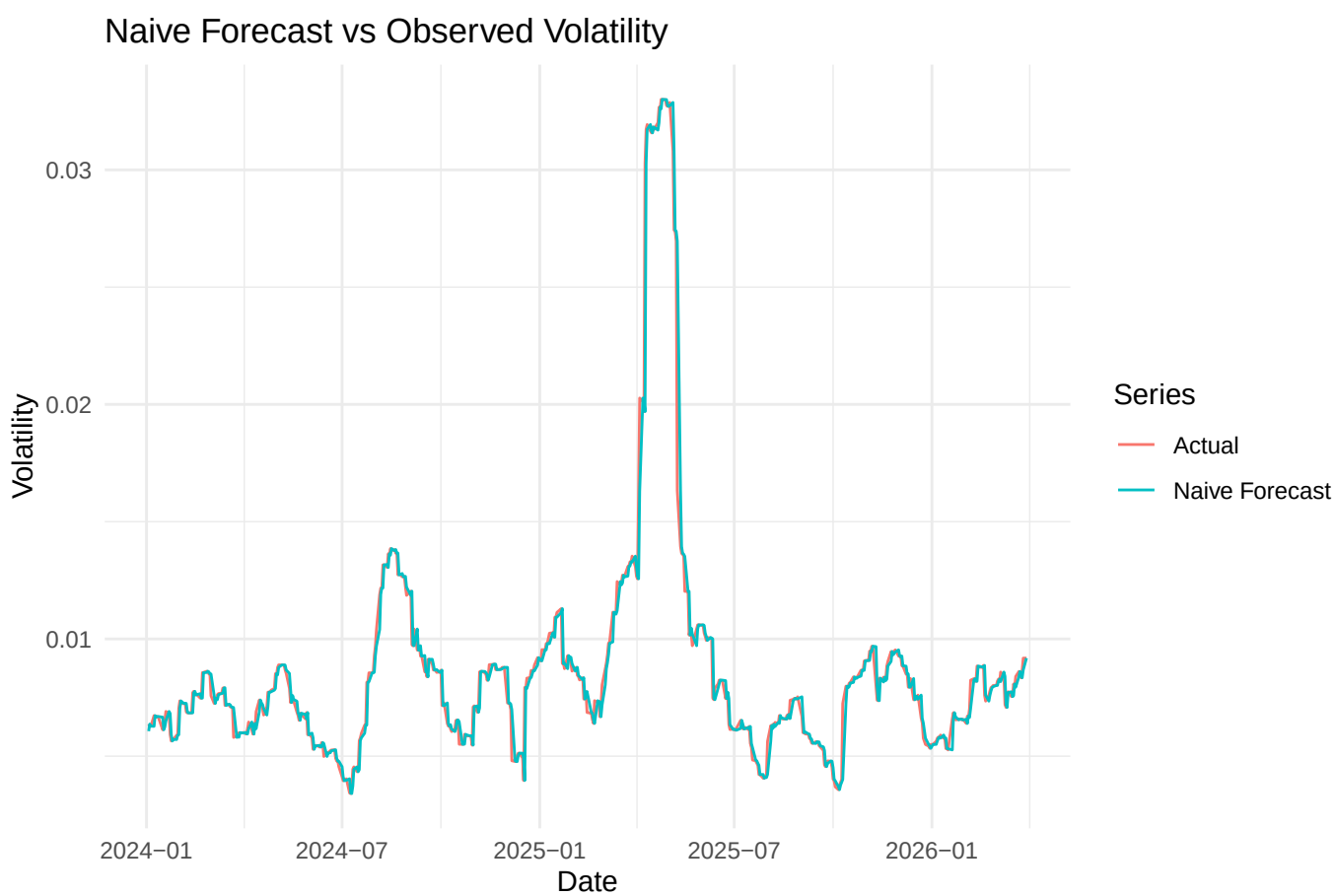

Figure 7: Updating one-step-ahead Naive forecast compared with observed volatility.

Under the updating one-step-ahead evaluation, the naive model achieved RMSE = 0.0009, MAE = 0.0003, and MAPE = 3.96.

4.2 ARIMA Model

The ARIMA model was fitted directly to the rolling volatility proxy using the `auto.arima()` procedure. Automatic model selection chose ARIMA(2,1,2) on the training set. This specification is consistent with the ACF/PACF evidence, which suggested strong short-lag dependence and an autoregressive component.

Compared with the naive benchmark, ARIMA attempts to capture linear dependence more flexibly through its autoregressive and moving-average structure. In practice, this can improve fit when the underlying dynamics are stable, but it may still struggle when the series contains abrupt changes in volatility level.

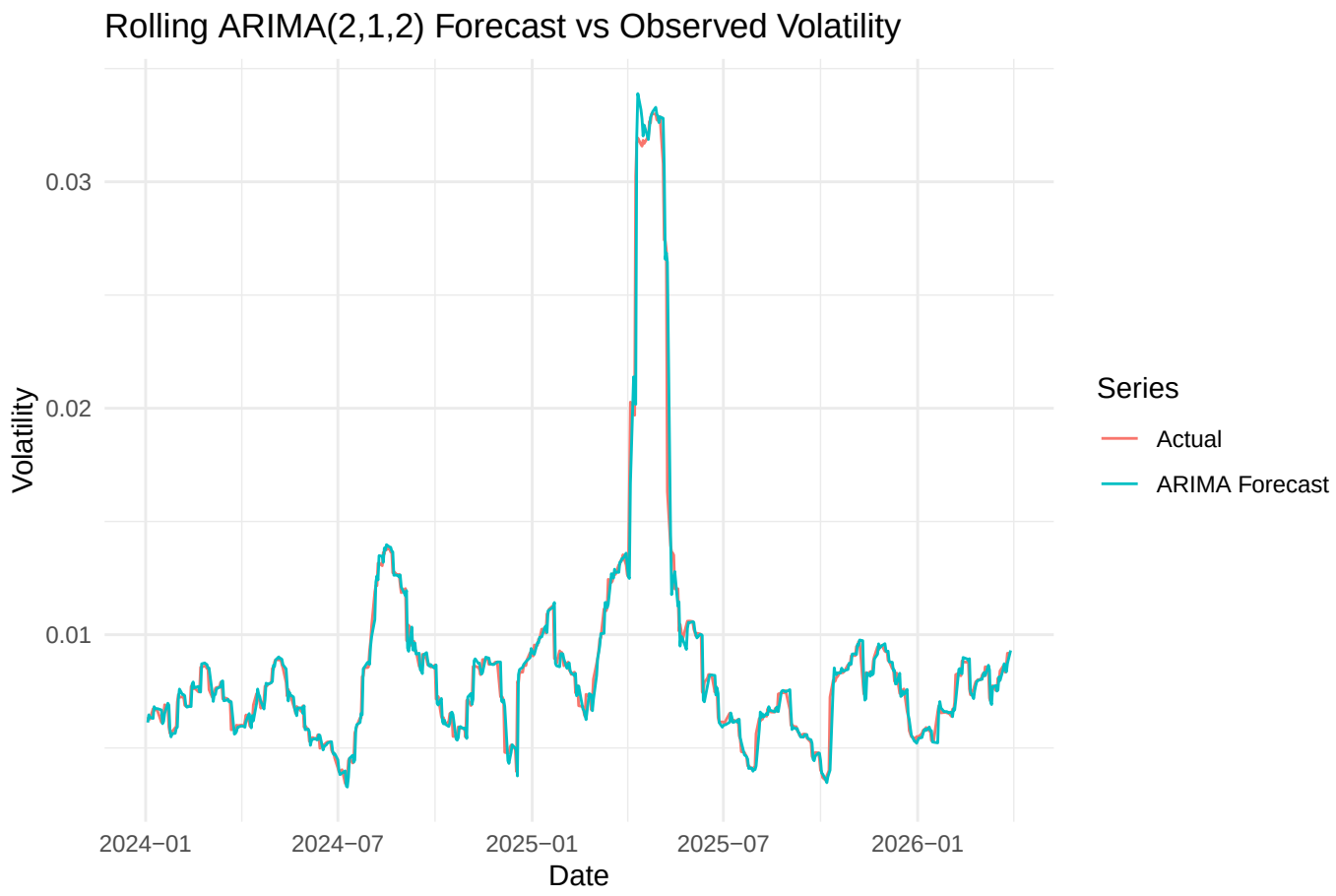

Figure 8: Updating one-step-ahead ARIMA forecast compared with observed volatility.

The ARIMA forecast follows the overall level of the series but still smooths through abrupt volatility changes. Its updating one-step-ahead accuracy was RMSE = 0.0008, MAE = 0.0004, and MAPE = 4.30.

ARIMA Residual Diagnostics

Residual diagnostics are useful for determining whether the ARIMA specification has removed most of the linear dependence in the training series. The residual plot, residual ACF, and residual histogram are shown below.

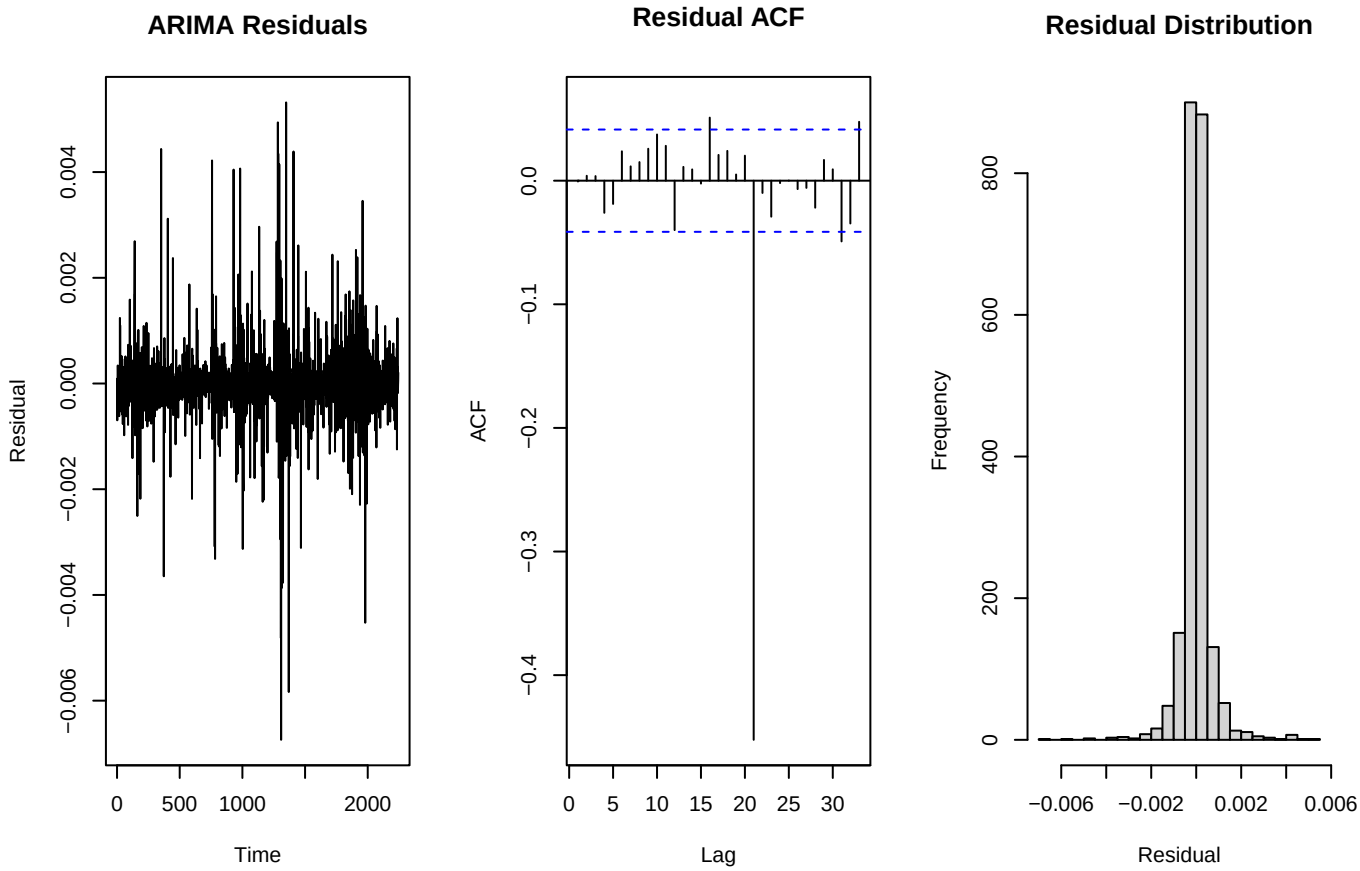

Figure 9: Residual diagnostics for the training-sample ARIMA model.

The residual diagnostics suggest that the ARIMA model captures part, but not all, of the dependence in the series. This is unsurprising because volatility data are not only persistent but also conditionally heteroskedastic, and ARIMA does not explicitly model time-varying variance.

4.3 ETS Model

The ETS model provides a state-space exponential smoothing benchmark. On the training set, the selected method was ETS(M,N,N). In this application, ETS acts mainly as a smooth level-based forecasting model rather than as a structural volatility model.

When ETS is used for a long single-origin forecast, the forecast path can appear nearly flat over the test period because the model is producing a conditional expectation from one fixed origin. For this reason, the ETS model is also evaluated using updating one-step-ahead forecasts, which provide a more interpretable view of how the model tracks the volatility series over time.

The ETS model is stable and often close to the naive benchmark, but it adapts gradually rather

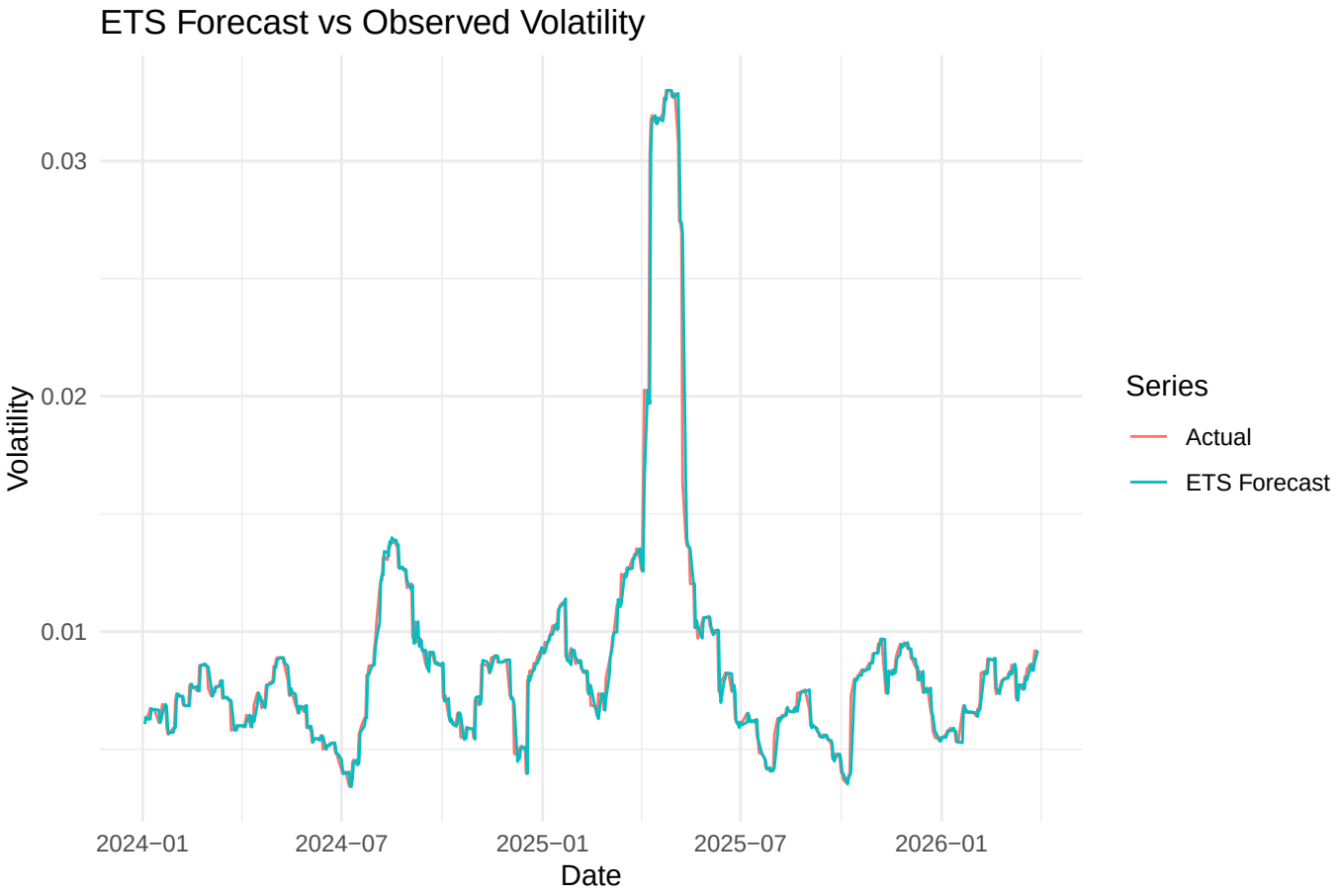

Figure 10: Updating one-step-ahead ETS forecast compared with observed volatility.

than abruptly. Its updating one-step-ahead accuracy was RMSE = 0.0009, MAE = 0.0003, and MAPE = 4.03.

4.4 NNAR Model

The neural network autoregression model, fitted using `nnetar()`, was included to assess whether nonlinear autoregressive structure could improve volatility forecasts. In principle, NNAR can capture nonlinear relationships missed by linear time series models. In practice, however, neural models often require careful tuning and can overfit when the underlying process is dominated by persistence rather than rich nonlinear patterns.

To reduce instability and keep computation manageable, the final report uses a modest NNAR specification and evaluates it under both the single-origin and updating one-step-ahead frameworks. The figure below shows the updating one-step-ahead path.

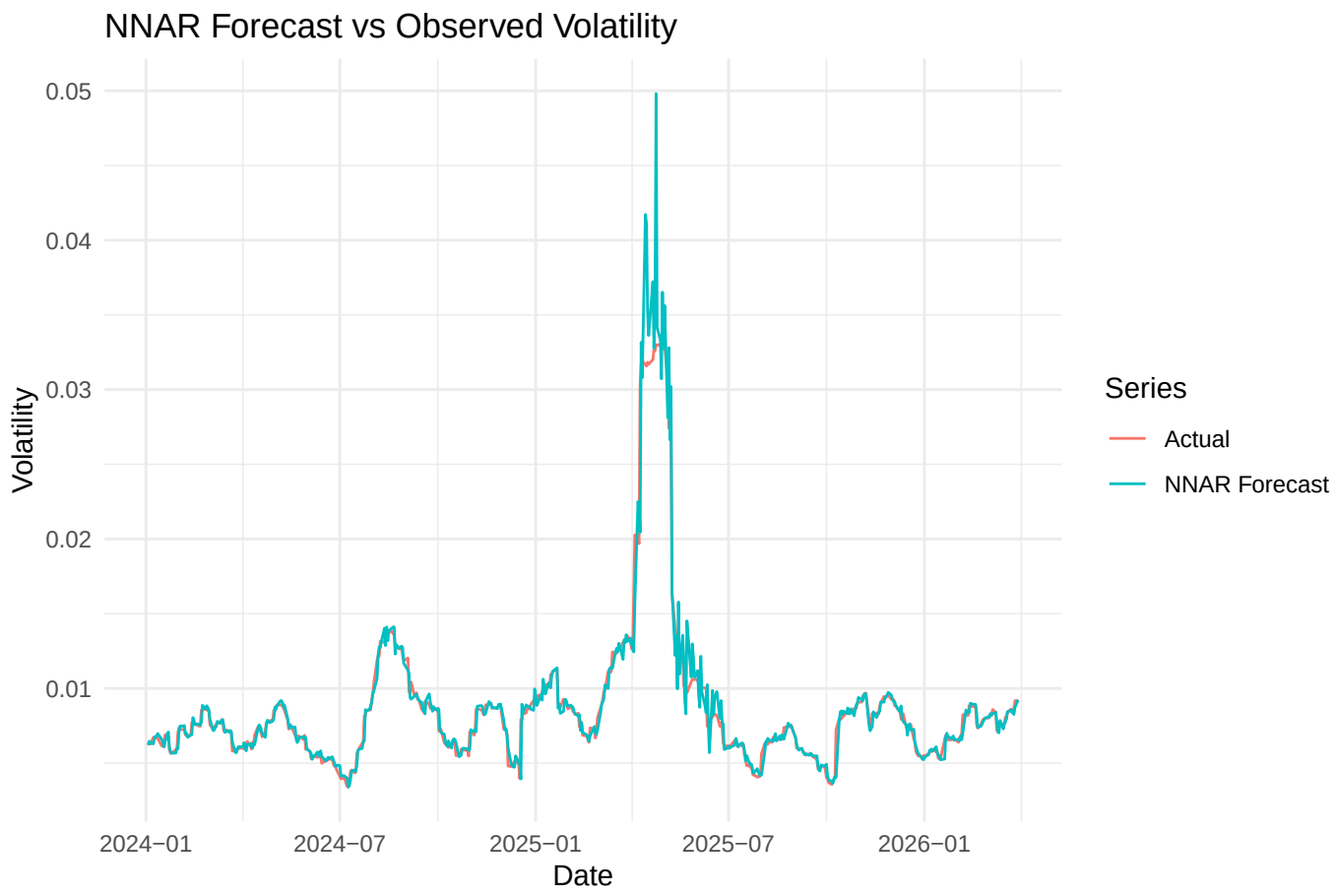

Figure 11: Updating one-step-ahead NNAR forecast compared with observed volatility.

The NNAR model is the least stable among the retained models. It sometimes tracks direction changes, but its path is generally less reliable and its forecast errors are larger. Its updating one-step-ahead accuracy was RMSE = 0.0013, MAE = 0.0005, and MAPE = 4.78.

4.5 GARCH Model

Because volatility clustering is the defining feature of this problem, the GARCH model is central to the analysis. Unlike the other models, which forecast the volatility proxy directly, GARCH is fitted to the return series and models the conditional variance recursively. This makes it especially appropriate for financial data where the variance changes through time.

The implementation uses two forecast constructions. For the single-origin benchmark, the conditional variance forecast is propagated forward analytically from the end of the training period. For the updating one-step-ahead evaluation, the forecast is updated recursively using the previous realized return and the previous conditional variance estimate. This approach aligns the GARCH forecast with the sequential forecasting setting used in the dynamic comparison.

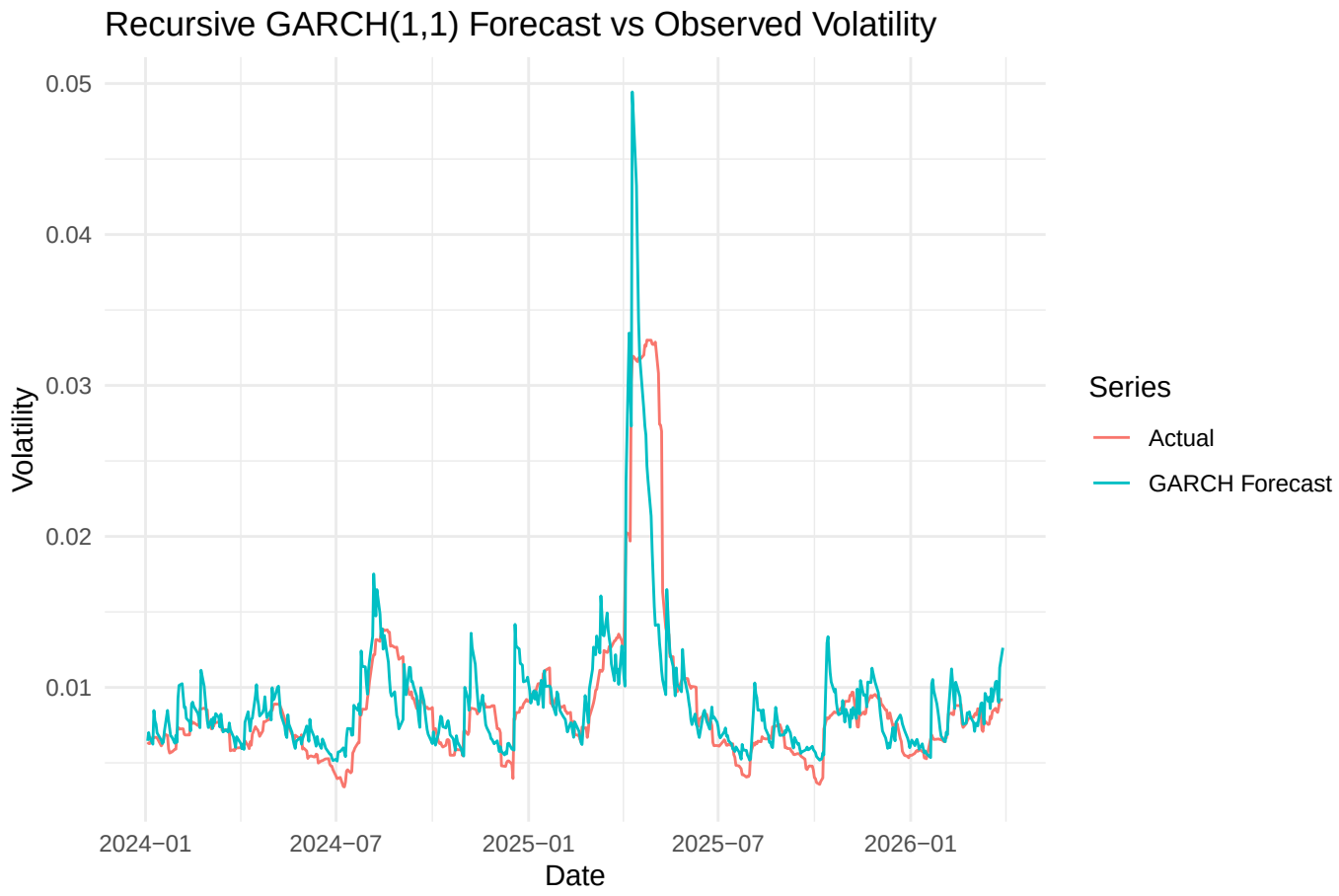

Figure 12: Recursive updating one-step-ahead GARCH(1,1) forecast compared with observed volatility.

The estimated GARCH parameters are shown below.

Table 4: Estimated GARCH(1,1) coefficients used in the recursive volatility update.

Parameter	Estimate
omega	0.000004
alpha	0.186014
beta	0.786091

The GARCH model is theoretically well matched to the financial interpretation of volatility because it models conditional variance directly. However, its forecast path is more reactive than the smoother 21-day rolling volatility proxy used as the evaluation target. As a result, GARCH is useful for interpreting volatility clustering, but it does not provide the best empirical accuracy in the updating one-step-ahead comparison. Its updating one-step-ahead accuracy was RMSE = 0.0030, MAE = 0.0017, and MAPE = 17.91.

5 Comparative Model Evaluation

The model-specific sections above describe the behavior of each forecast path. This section compares the models side by side using both the single-origin benchmark and the updating one-step-ahead evaluation.

5.1 Single-Origin Benchmark Results

The first comparison table retains the single-origin design used in the original presentation workflow. Each model is trained once on the training data and then forecast across the entire test horizon. This table is useful for continuity with the slide deck and for evaluating how models behave when a long-horizon forecast is required from one fixed origin.

Table 5: Single-origin benchmark forecast accuracy across competing models.

Model	RMSE	MAE	MAPE
Naive	0.0057	0.0030	26.7471
ARIMA	0.0055	0.0028	24.9858
GARCH	0.0057	0.0044	59.6264
ETS	0.0057	0.0030	26.7472
NNAR	0.0057	0.0030	25.9951

Under the single-origin benchmark, the best RMSE was obtained by ARIMA, with an RMSE of 0.0055. This benchmark is the most directly comparable to the class presentation, but the full-horizon forecast paths for simple models should be interpreted cautiously because they are not meant to visually track each test observation in real time.

Benchmark Error Heatmap

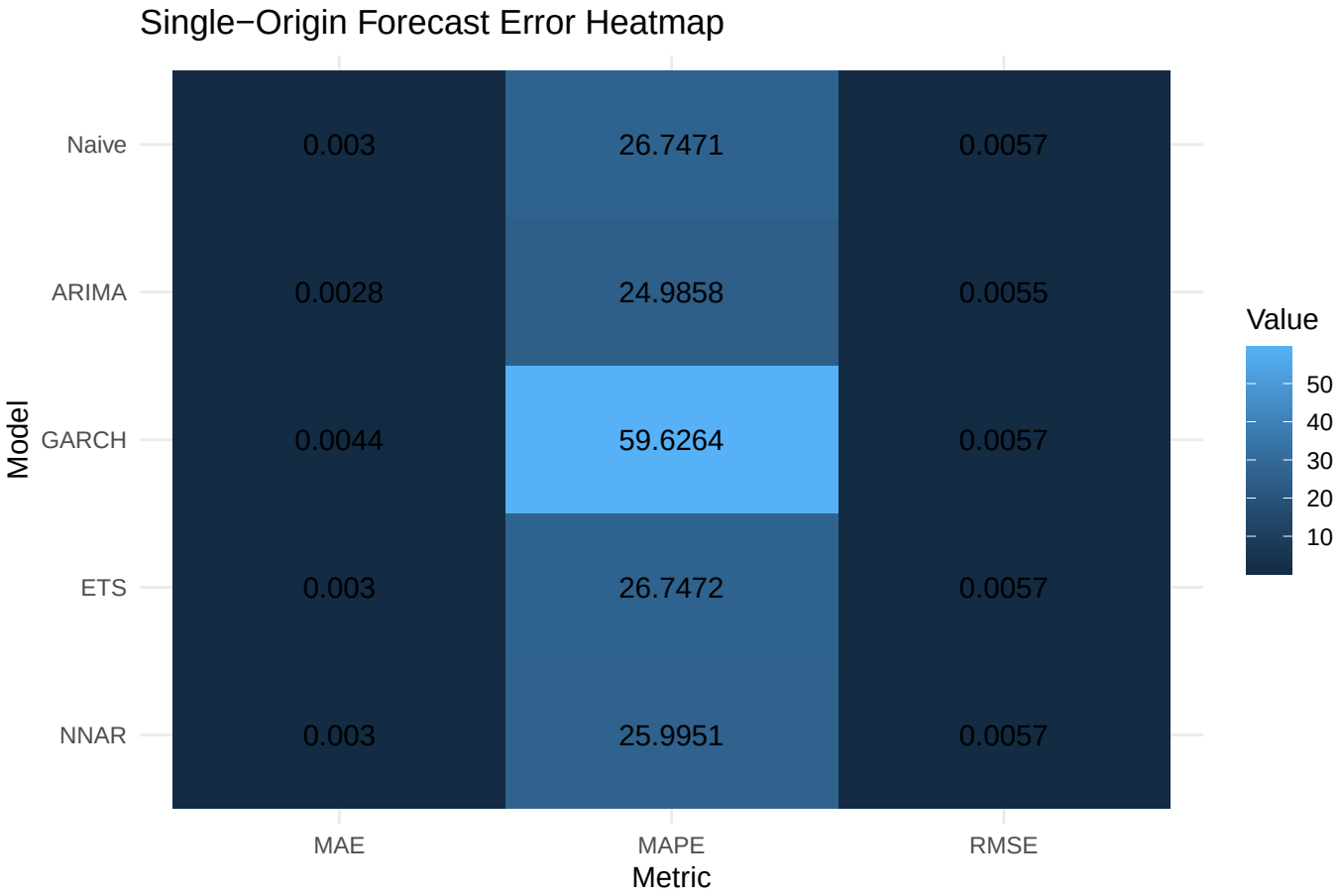

Figure 13: Heatmap of single-origin forecast errors across models and metrics.

Benchmark Ranking Heatmap

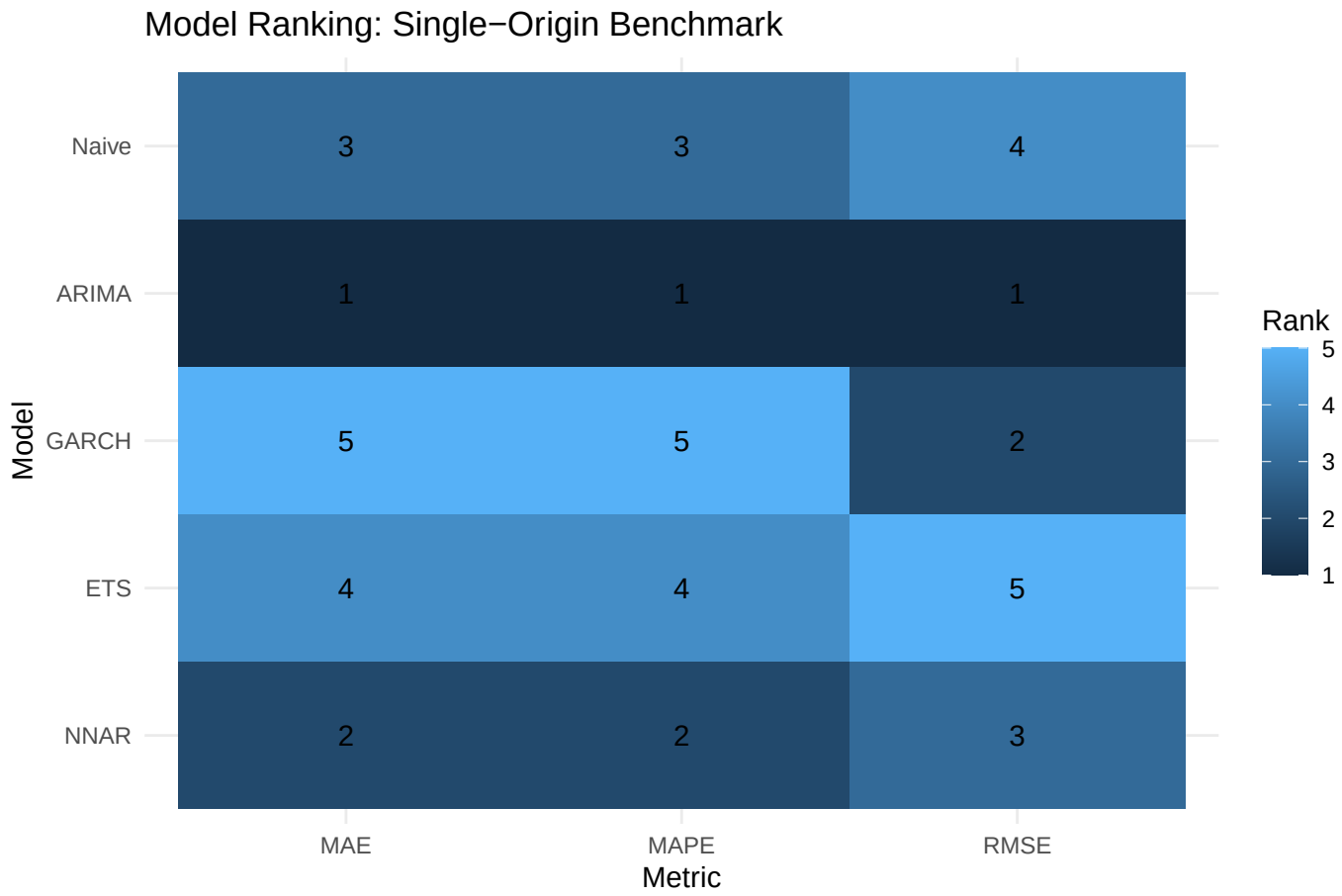

Figure 14: Model ranking under the single-origin benchmark (1 = best).

The benchmark ranking heatmap shows whether the relative ordering of the models is stable across RMSE, MAE, and MAPE. A consistent ranking across metrics indicates that the preferred model is not winning only because of one particular error measure.

5.2 Updating One-Step-Ahead Results

The second table reports the updating one-step-ahead accuracy measures. This evaluation is the preferred basis for interpreting the plotted forecast paths because the models are updated as new observations become available.

Table 6: Updating one-step-ahead forecast accuracy across competing models.

Model	RMSE	MAE	MAPE
Naive	0.0009	0.0003	3.9554
ARIMA	0.0008	0.0004	4.3027
GARCH	0.0030	0.0017	17.9116

Model	RMSE	MAE	MAPE
ETS	0.0009	0.0003	4.0294
NNAR	0.0013	0.0005	4.7807

Under the updating one-step-ahead evaluation, the lowest RMSE was obtained by ARIMA, with an RMSE of 0.0008. In quantitative terms, ARIMA reduces RMSE by approximately 11% relative to the Naive benchmark (0.0008 vs 0.0009), indicating a modest but consistent improvement in capturing short-term volatility dynamics. However, the ranking is metric-dependent: ARIMA performs best by RMSE, while the Naive benchmark performs best by MAE and MAPE. This result suggests that the volatility series is highly persistent, so simple persistence-based forecasts remain difficult to beat even when more flexible models are included. This updating framework is the preferred operational comparison because it incorporates new information sequentially over time.

Updating One-Step Ranking Heatmap

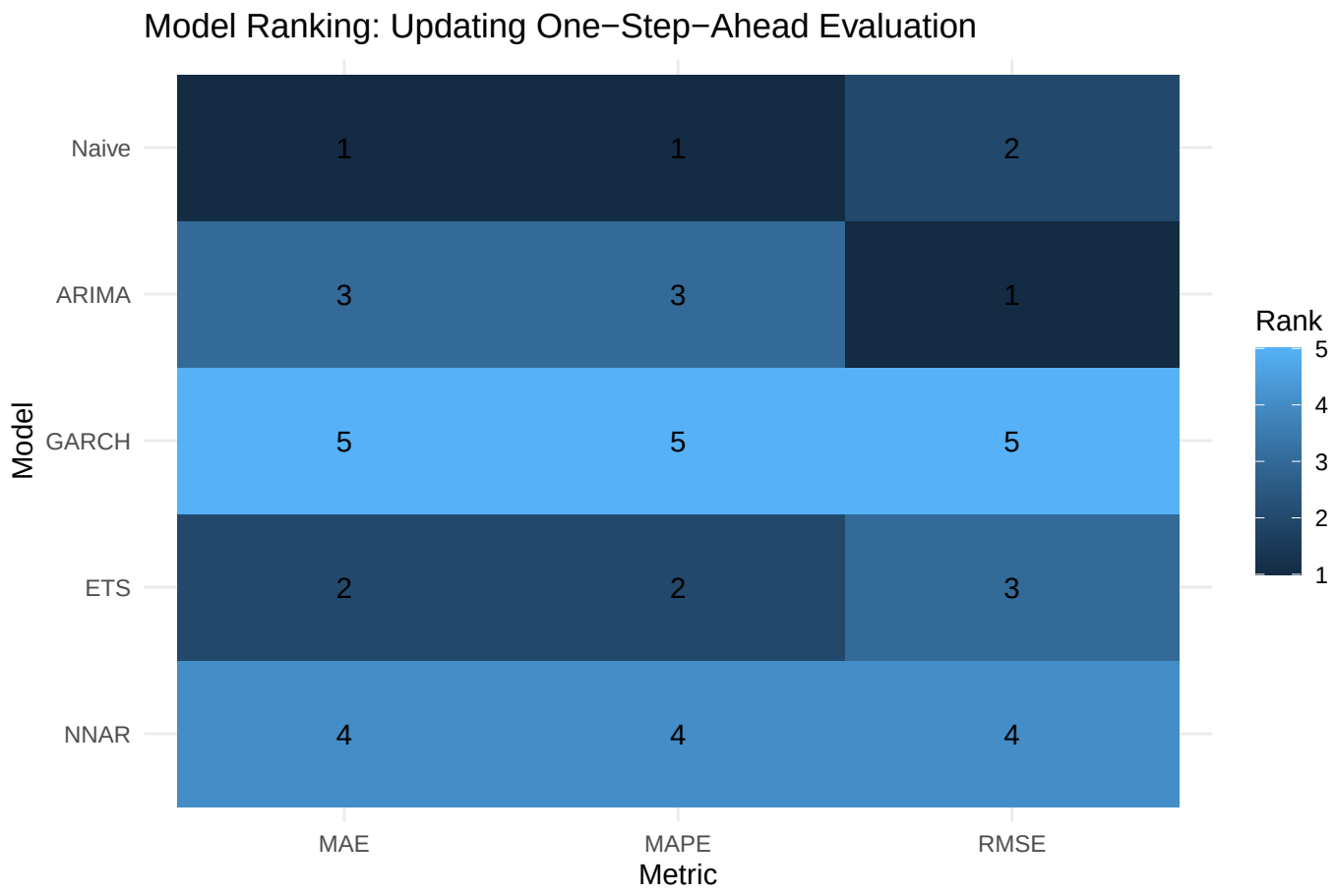

Figure 15: Model ranking under updating one-step-ahead evaluation (1 = best).

Updating One-Step Error Heatmap

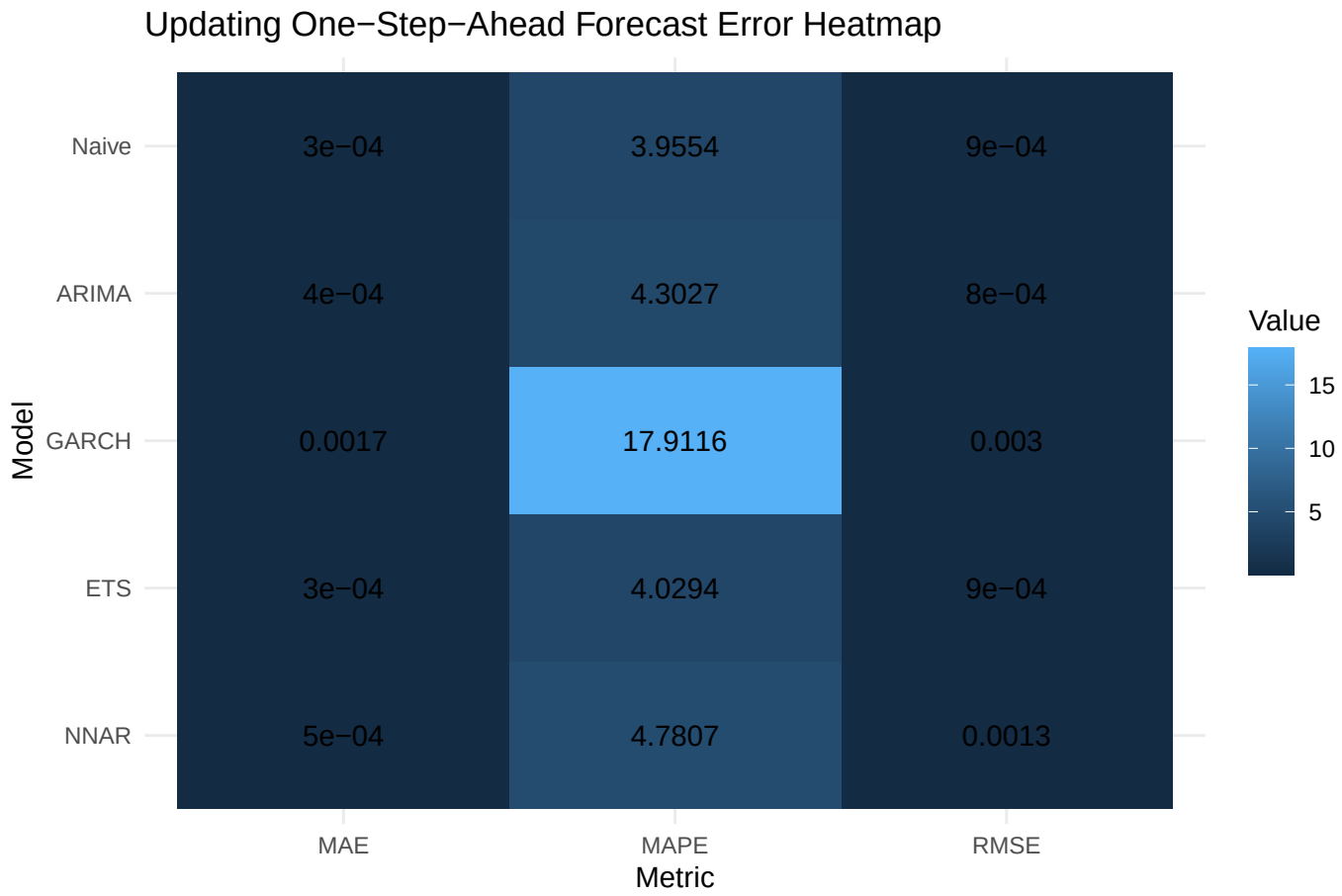

Figure 16: Heatmap of updating one-step-ahead forecast errors across models and metrics.

5.3 All-Model Dynamic Forecast Comparison

The figure below places all updating one-step-ahead forecasts on the same chart. This view is intentionally dense, but it is useful for seeing the broad differences in responsiveness, smoothness, and overall tracking behavior across all models.

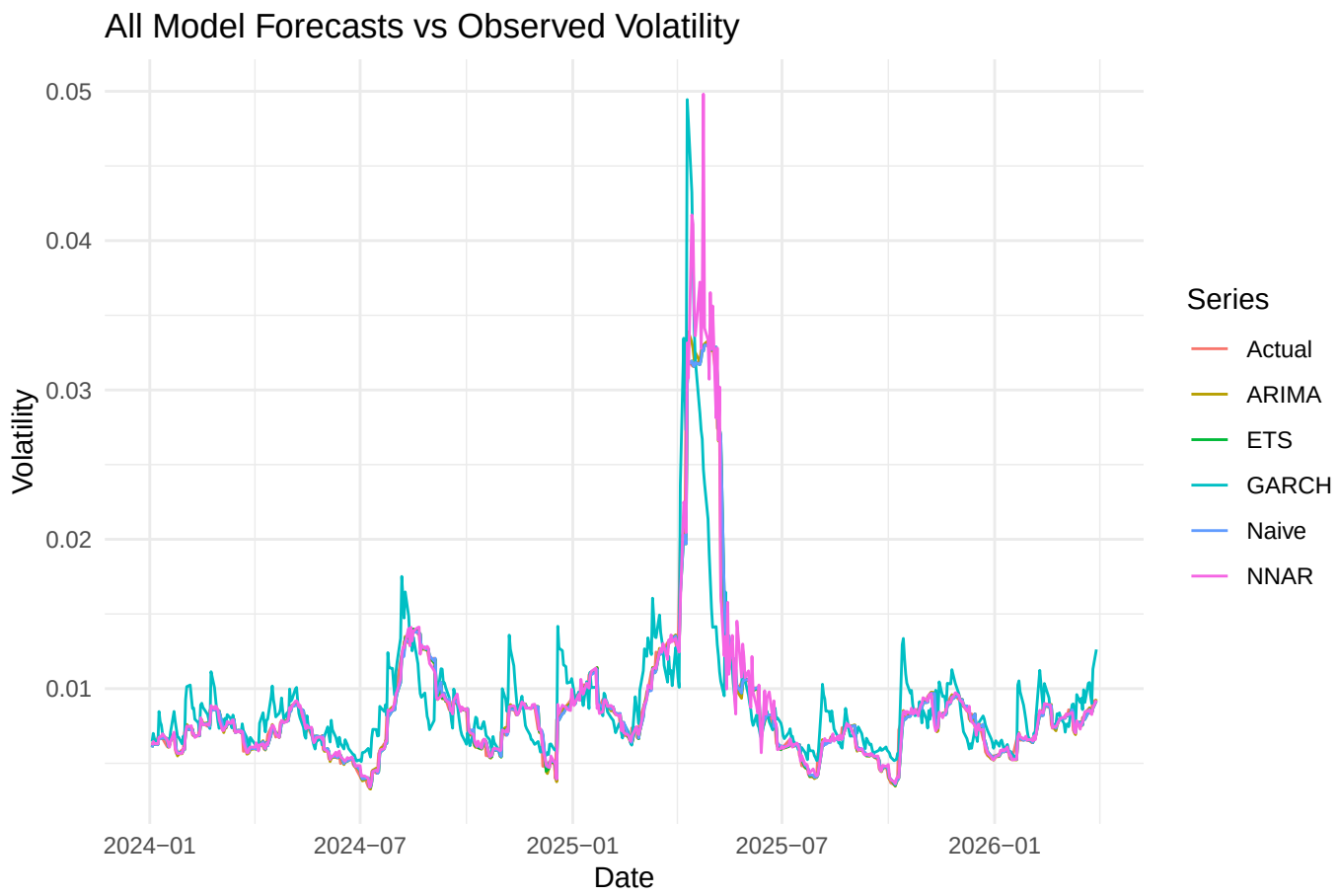

Figure 17: Updating one-step-ahead forecast paths for all retained models compared with observed volatility.

The all-model comparison reinforces several qualitative patterns. The naive and ETS models are stable and persistent. ARIMA is somewhat more flexible than these simple benchmarks but still mostly smooths over abrupt changes. NNAR appears less reliable and more erratic. The GARCH path is the most interpretable from a volatility perspective because it adapts to shocks while preserving mean-reverting variance behavior.

5.4 Conceptual and Benchmark Model Comparison

The following figure focuses on three interpretable benchmark models: GARCH, ETS, and the Naive benchmark. This comparison highlights the contrast between a variance-based financial model and two simpler persistence-based alternatives. These are not necessarily the top three models by every accuracy metric; rather, they are useful for interpreting the tradeoff between theoretical motivation and empirical forecast accuracy.

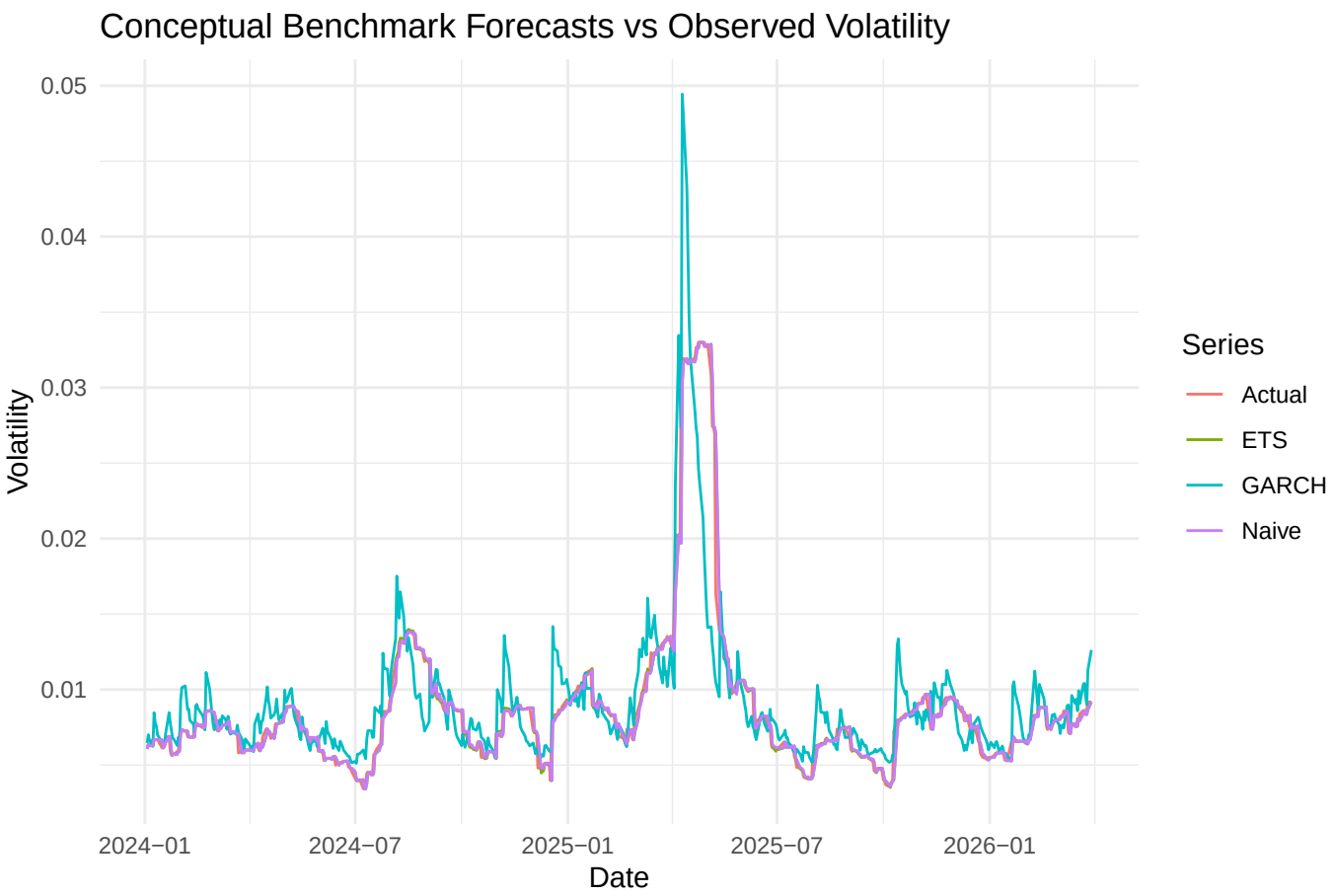

Figure 18: Observed volatility compared with selected conceptual and benchmark forecast paths.

Among these models, GARCH is the most theoretically motivated because it explicitly models conditional heteroskedasticity. However, the Naive and ETS forecasts are more competitive empirically because the 21-day rolling volatility target is highly persistent and relatively smooth. This comparison therefore shows that theoretical suitability does not automatically guarantee the lowest forecast error for a constructed rolling-volatility proxy.

6 Discussion

The empirical results answer the project's research questions in a reasonably clear way. First, short-term SPY volatility is forecastable to some extent. The volatility proxy exhibits strong persistence, clustered high-variance episodes, and a pronounced dependence structure, all of which give forecasting models exploitable signal. Second, more advanced models do not automatically outperform simple baselines. In both the benchmark and updating one-step-ahead evaluations, the simple models remain competitive because the volatility series is highly persistent.

Third, the best model depends somewhat on the evaluation design, but the overall interpretation is stable. The GARCH model is the most appropriate model conceptually because it directly represents conditional heteroskedasticity in returns. Empirically, however, it does not dominate the updating one-step-ahead accuracy table. ARIMA has the lowest RMSE, while the Naive benchmark has the

strongest MAE and MAPE. This difference suggests that the evaluation target matters: the 21-day rolling volatility proxy is smoother than the one-period conditional volatility estimated by GARCH, so persistence-based models can perform very well even without explicitly modeling return variance.

The comparison also illustrates why plotting choices matter. A direct multi-step forecast from a simple level model over a long test window can create the misleading impression that the model is "wrong" simply because the plotted line looks nearly constant. In reality, that path is a valid long-horizon conditional expectation from one origin. For tracking the evolution of volatility through time, however, the updating one-step-ahead evaluation is more informative. This is why the final report now separates benchmark accuracy from dynamic path behavior.

The NNAR results are also informative even though the model is not among the strongest performers. The neural network benchmark was included because nonlinear structure is plausible in financial volatility data. However, the results suggest that in this application the volatility process is dominated by persistence and conditional variance dynamics rather than by nonlinear patterns that can be easily exploited by a relatively small autoregressive neural network. This is consistent with the presentation conclusion that NNAR tended to overfit and generalize poorly.

7 Limitations and Future Work

Several limitations should be acknowledged. First, the target series is a rolling volatility proxy rather than a direct realized volatility measure. This makes the problem interpretable and computationally convenient, but it is still a constructed series rather than latent volatility itself. Second, the analysis focuses on a single market proxy, SPY, and therefore does not capture cross-asset variation. Third, no external predictors were included, even though variables such as VIX, interest rates, or macroeconomic announcements could plausibly improve short-term risk forecasts.

A fourth limitation concerns the evaluation design itself. Although the updating one-step-ahead framework is more realistic than a purely static holdout plot, the models were not re-selected from scratch in a full rolling-window or rolling-origin backtesting exercise. For example, the ARIMA order was selected once and then applied repeatedly, and the neural model was intentionally kept modest to avoid excessive computational cost. These are reasonable compromises for a class project, but they are still simplifications relative to a production forecasting system.

Future work could extend the report in several directions. One natural extension is to compare alternative volatility models such as EGARCH, GJR-GARCH, or stochastic volatility models. Another is to introduce exogenous predictors, especially market-implied volatility measures, macroeconomic variables, or sector-specific information. A third direction is to evaluate rolling one-step-ahead forecasts with repeated full model re-estimation, which would provide an even stronger robustness check. Finally, the analysis could be repeated across multiple assets to determine whether the same model ranking holds outside a broad-market ETF.

8 Summary and Conclusions

This project examined whether short-term stock market volatility can be forecasted using standard time series methods, using SPY daily returns and a 21-day rolling volatility proxy. The data showed strong persistence, volatility clustering, and substantial changes in risk over time. These features motivated a comparison among Naive, ARIMA, ETS, NNAR, and GARCH approaches.

The final results show that volatility is forecastable, but the models differ in both interpretability and performance. Simple persistence-based benchmarks remain competitive, which indicates that much of the information in the series lies in its own recent history. This pattern arises because the volatility proxy is a smoothed, highly persistent series, meaning that most predictive signal is contained in recent observations. As a result, flexible models such as NNAR or GARCH do not gain a substantial advantage unless they capture additional structure beyond persistence. At the same time, the GARCH framework remains especially compelling because it is built to model conditional variance directly and therefore aligns most naturally with the underlying financial problem.

A central methodological feature of the final analysis is the distinction between single-origin forecasts and updating one-step-ahead forecasts. The single-origin benchmark evaluates long-horizon performance from one fixed training endpoint, while the updating framework better represents an operational monitoring setting in which new observations become available sequentially. This distinction clarifies why simple persistence-based models can look visually flat in long-horizon forecasts but remain competitive in one-step-ahead accuracy.

9 AI Use Statement

AI tools were used to improve the organization of the report, refine written English, suggest revisions to the R Markdown structure, and help diagnose earlier model-implementation issues. All analyses, code execution, model checking, and final interpretation were reviewed and validated by the authors.